

Derive the loss from the likelihood

The loss is not an arbitrary penalty: it is the negative log-score of a predictive distribution.

Distribution assumption → per-example NLL → familiar training loss.

1 • The universal derivation

Conditional independence gives

$$p(\mathcal{D}|\theta) = \prod_i p_{\theta}(y_i|\mathbf{x}_i).$$

Taking logs and changing maximization to minimization gives

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} - \sum_i \log p_{\theta}(y_i|\mathbf{x}_i).$$

Define one-example loss

$$\ell_{i(\theta)} = -\log p_{\theta}(y_i|\mathbf{x}_i).$$

The dataset objective is the sum or mean of these contributions.

2 • Gaussian likelihood gives squared error

Assume

$$Y_i|\mathbf{x}_i \sim \mathcal{N}(\mu_i, \sigma^2), \quad \mu_i = f_{\theta}(\mathbf{x}_i).$$

Then

$$-\log p(y_i|\mathbf{x}_i) = \frac{1}{2\sigma^2}(y_i - \mu_i)^2 + \log \sigma + C.$$

With fixed σ ,

$$\hat{\theta}_{\text{MLE}} = \arg \min_{\theta} \sum_i (y_i - f_{\theta}(\mathbf{x}_i))^2.$$

MSE is Gaussian NLL up to positive scale and additive constants. Learning σ requires keeping the $\log \sigma$ term.

3 • Residual model chooses the penalty

Let $r = y - \hat{y}$.

Noise	NLL in r	Effect
Gaussian	r^2	large errors dominate
Laplace	$ r $	constant influence
Student- t	$\log\left(1 + \frac{r^2}{\nu s^2}\right)$	heavy-tail robustness
Uniform	0 inside; ∞ outside	hard support

Constants and positive scale factors are omitted in the compact table.

4 • Bernoulli likelihood gives BCE

For $y \in \{0, 1\}$ and predicted $p = P(Y = 1|\mathbf{x})$,

$$p(y|\mathbf{x}) = p^{y(1-p)^{1-y}}.$$

Therefore

$$\ell_{\text{BCE}} = -[y \log p + (1 - y) \log(1 - p)].$$

If $y = 1$, $p = 0.8$ gives $\ell \approx 0.22$ while $p = 0.2$ gives $\ell \approx 1.61$.

Use a logit $z \in \mathbb{R}$ and a fused BCE-with-logits operation for numerical stability.

5 • Categorical likelihood gives cross-entropy

For mutually exclusive classes, logits z define

$$p_k = \frac{\exp z_k}{\sum_j \exp z_j}.$$

If the observed class is c ,

$$\ell_{\text{CE}} = -\log p_c = -z_c + \log \sum_j \exp z_j.$$

Cross-entropy depends on the probability assigned to the true class. Confident mistakes are expensive because $-\log p_c \rightarrow \infty$ as $p_c \rightarrow 0$.

6 • Match observation, output and loss

Target	Model output	NLL
real binary	μ ; optionally $\sigma > 0$ one unconstrained	Gaussian / robust Bernoulli BCE
one of K	logit K unconstrained	categorical CE
count	logits positive rate λ	Poisson NLL

Before optimizing a loss, say aloud which random variable and distribution make it the right score.

From MLE to MAP and regularization

Likelihood asks for data fit; a prior adds an explicit preference among plausible parameter settings.

1 • Bayes combines data and prior

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})}.$$

likelihood	how the fixed data score each θ
prior	pre-data preference over θ
posterior	updated distribution after observing data
evidence	normalizer; independent of θ

2 • MAP is loss plus regularizer

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} p(\theta|\mathcal{D}).$$

Dropping the evidence and taking negative logs,

$$\hat{\theta}_{\text{MAP}} = \arg \min_{\theta} \left[\underbrace{-\log p(\mathcal{D}|\theta)}_{\text{NLL}} + \underbrace{-\log p(\theta)}_{\text{prior penalty}} \right].$$

MLE uses only the likelihood.

MAP returns one parameter vector at the posterior mode.

3 • Gaussian prior gives L_2

If $\theta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$,

$$-\log p(\theta) = \frac{1}{2\tau^2} \|\theta\|_2^2 + C.$$

Hence

$$\mathcal{L}_{\text{MAP}} = \text{NLL} + \lambda \|\theta\|_2^2, \quad \lambda = \frac{1}{2\tau^2}.$$

Smaller τ means a narrower prior, larger λ , and stronger shrinkage toward zero.

4 • Laplace prior gives L_1

For independent zero-centred Laplace parameters with scale a ,

$$-\log p(\theta) = \frac{1}{a} \|\theta\|_1 + C.$$

The corner at zero can threshold coefficients exactly to zero. L_2 smoothly shrinks; L_1 can create sparsity.

5 • One-dimensional MAP shrinkage

Approximate the likelihood by

$$p(\mathcal{D}|\theta) \propto \exp\left(-\frac{(\theta-z)^2}{2s^2}\right)$$

with $z = \hat{\theta}_{\text{MLE}}$, and use $\theta \sim \mathcal{N}(0, \tau^2)$. Minimizing the two quadratic terms gives

$$\hat{\theta}_{\text{MAP}} = \underbrace{\frac{\tau^2}{\tau^2 + s^2}}_{\alpha \in (0,1)} z.$$

If $z = 0.6$ and $s = \tau$, then $\hat{\theta}_{\text{MAP}} = 0.3$.

6 • Scaling: keep the statistical meaning

The exact MAP objective uses **summed** NLL plus the full prior penalty:

$$\sum_i \ell_{i(\theta)} + \lambda_{\text{sum}} R(\theta).$$

If code uses mean NLL, divide the whole objective by n :

$$\frac{1}{n} \sum_i \ell_{i(\theta)} + \lambda_{\text{mean}} R(\theta), \quad \lambda_{\text{mean}} = \frac{\lambda_{\text{sum}}}{n}.$$

Changing batch size should not silently change the intended regularization strength. Document whether a library returns a sum or mean.

7 • What stronger priors do

More data	likelihood sharpens; MAP often approaches MLE
Narrower prior	posterior mode moves farther toward preferred values
Correlated prior	preferred directions become elliptical, not axis-aligned
Prior centre m	shrinkage is toward m , not automatically toward zero

8 • Final audit

- ✓ Each loss is tied to an explicit likelihood.
- ✓ Constants are dropped only when they do not affect the optimized parameters.
- ✓ Logits and constrained distribution parameters are not confused.
- ✓ Prior family and scale are stated.
- ✓ Sum-versus-mean scaling is explicit.

Likelihood gives the data loss. Prior gives the regularizer. Their posterior mode gives MAP.